**Clinical Study Summary Report (CSSR)**
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## 1. Administrative & Study Identification

**Full Study Title:** Long-Term Safety Outcomes and First-Line Treatment Patterns in Patients With Non-Small Cell Lung Cancer and Programmed Death-1 (PD-L1) <1% **Official Title:** Real-World Long-Term Safety Outcomes and First-Line Treatment Patterns in Patients With Non-Small Cell Lung Cancer and Pd-L1 <1% in the Florida Cancer Specialists & Research Institute Database **Short Title/Acronym:** NSCLC PD-L1 <1% RWE Safety Study
**Protocol Number:** [Internal Study ID]
**NCT Number:** NCT07215962
**Sponsor Name:** Bristol Myers Squibb **Investigational Product:** Nivolumab (Opdivo) + Ipilimumab (Yervoy) $\pm$ Platinum-based chemotherapy **Phase of Development:** Not Applicable
**Trial Status:** ACTIVE_NOT_RECRUITING **Medical Monitor:** Bristol Myers Squibb Medical Director

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## 2. Study Synopsis & Rationale

**2.1 Study Objective Primary Objective:** To assess the treatment-related adverse events and associated healthcare resource use in programmed death ligand 1 (PD-L1) negative individuals diagnosed with advanced/metastatic non-small cell lung cancer (NSCLC) who received first-line therapy. **Secondary Objectives:** To characterize first-line treatment patterns, overall survival, and long-term tolerability across different treatment cohorts in a real-world setting.

**2.2 Background & Rationale** To address the gap in real-world evidence regarding the long-term safety and treatment patterns of dual immuno-oncology therapies (nivolumab + ipilimumab) with or without chemotherapy in patients with advanced NSCLC who are PD-L1 negative (<1%). This population historically has had limited immunotherapy efficacy when used as monotherapy, necessitating combination strategies whose real-world healthcare impact and tolerability require further characterization.

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## 3. Study Design & Methodology

**3.1 Design Framework Study Type:** Observational, Retrospective/ Prospective Cohort Study **Control Type:** Active Comparator (Standard of Care cohorts) **Blinding:** Open-label (Unblinded) **Randomization:** N/A (Real-World Evidence)

**3.2 Planned Enrollment & Duration Target $N$:** 300 **Number of Sites:** Multiple US/Global healthcare databases and clinical registries (including Florida Cancer Specialists & Research Institute Database) **Estimated Study Start:** [Retrospective index dating] **Estimated Primary Completion:** October 2025

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## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Are $\ge 18$ years of age at the index date. 2. Have a confirmed diagnosis of advanced/metastatic non-small cell lung cancer (NSCLC) (stage IIIB-IV) (squamous and non-squamous). 3. Have PD-L1 < 1% level as reported. 4. Received one of the following 1L treatments: * Cohort 1: nivolumab + ipilimumab * Cohort 2: nivolumab + ipilimumab + platinum-based chemotherapy * Cohort 3: Immuno-oncology (IO)-based therapy (excluding nivolumab-based regimens) with chemotherapy * Cohort 4: Dual-IO with chemotherapy (eg. tremelimumab-actl + durvalumab + carboplatin + albumin-bound paclitaxel, tremelimumab-actl + durvalumab + [carboplatin or cisplatin] + gemcitabine, tremelimumab-actl + durvalumab + carboplatin + albumin-bound paclitaxel, tremelimumab-actl + durvalumab + [carboplatin or cisplatin] + pemetrexed) 5. Have $\ge 6$ months of documented post-index (follow-up) period after the index date - Participants who die within 6 months of follow-up will be included.

**4.2 Exclusion Criteria** 1. Have positive or unknown EGFR or ALK mutation before the index date. 2. Have a gap of > 120 days between metastatic NSCLC diagnosis and index date. 3. Were included in a clinical trial for 1L therapy. 4. Enter a hospice within 6 months of the follow-up will be excluded. 5. Have other concurrent primary cancer diagnoses.

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## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen Dosage/Frequency:** Dosing as per standard of care and real-world prescribing patterns. Patients are divided into cohorts based on 1L treatments: * Cohort 1: nivolumab + ipilimumab * Cohort 2: nivolumab + ipilimumab + platinum-based chemotherapy * Cohort 3: Immuno-oncology (IO)-based therapy (excluding nivolumab-based regimens) with chemotherapy * Cohort 4: Dual-IO with chemotherapy **Route of Administration:** Intravenous (IV). **Duration of Treatment:** Treatment continued until disease progression, unacceptable toxicity, or maximum allowed duration per standard clinical practice.

**5.2 Concomitant & Prohibited Therapy Permitted:** Routine supportive care medications, anti-emetics, and symptom management as prescribed in real-world practice. **Prohibited:** Concurrent investigational agents (excluded from main cohorts).

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## 6. Study Timeline & Visit Schedule

*(Note: As an observational study, "visits" correlate to standard clinical encounters captured in health records)* | Visit ID | Window | Key Activities/Procedures | | :--- | :--- | :--- | | **Index Date** | Day 0 | Initiation of first-line therapy, baseline demographics, and tumor characteristics. | | **Follow-up** | Real-world intervals | Monitoring of Treatment-Related Adverse Events (TRAEs), healthcare resource utilization. | | **End of Study** | Data cutoff | Final extraction of Overall Survival (OS) and Time to Next Treatment (TTNT) records. |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoints** 1. Incidence of treatment-related adverse events. 2. Participant baseline clinical characteristics. 3. Time to onset of treatment-related adverse events.

**7.2 Secondary & Exploratory Endpoints** 1. Healthcare Resource Utilization (HCRU) associated with treatment-related adverse event.

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Extracted retrospectively/ prospectively from clinical notes, ICD/CPT diagnosis codes, and pharmacy claims. **Serious Adverse Events (SAEs):** Identified via emergency room visits and inpatient hospitalization records associated with immune-mediated toxicities. **Data Monitoring Committee (DMC):** Managed by the Sponsor's internal Real-World Data and Epidemiology teams.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** All eligible patients who initiated first-line therapy and met the inclusion criteria (Full Analysis Set). **Sample Size Justification:** Target enrollment of 300 patients based on maximum available patient records meeting the study criteria within the designated database timeframe. **Handling of Missing Data:** Multiple Imputation (MI) for missing baseline covariates, or complete case analysis depending on the specific endpoint modeling.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Pharmacoepidemiology Practices (GPP) and applicable patient privacy regulations (e.g., HIPAA). **Monitoring Plan:** Programmatic data quality checks, source data verification for a random sample of abstracted charts. **Audit Trail:** Secure, encrypted data management platforms with full audit logs of queries and data extraction scripts.

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## 11. Study Identifiers & Governance

**Primary ID:** N/A **Registry Number:** NCT07215962 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** Long-Term Safety Outcomes and First-Line Treatment Patterns in Patients With Non-Small Cell Lung Cancer and Programmed Death-1 (PD-L1) <1% **Official Regulatory Title:** Real-World Long-Term Safety Outcomes and First-Line Treatment Patterns in Patients With Non-Small Cell Lung Cancer and Pd-L1 <1% in the Florida Cancer Specialists & Research Institute Database

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## 12. Clinical Framework (Oncology/NSCLC Specific Adapted)

**Primary Condition:** Non-small Cell Lung Cancer (NSCLC) / Non-Small Cell Lung Carcinoma **Diagnosis Threshold:** PD-L1 expression < 1% (PD-L1 negative), Stage IIIB-IV **Medical Methodology:** First-Line Immuno-Oncology Combination Therapy **Optimization Target:** Disease control and long-term tolerability

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## 13. Study Procedures & Methodology

**Management Pathway:** Standard Oncology Care Pathway for NSCLC **Education Protocol (HCP):** N/A (Observational Real-World Data collection) **Education Protocol (Patient):** N/A (Routine clinical care) **Quality Audit Mechanism:** EHR data abstraction validation & Claims analysis tracking

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## 14. Observation & Follow-Up Schedule

**Total Duration:** Follow-up until death, loss to follow-up, or end of data availability (minimum 6 months) **Onsite Visit Cadence:** Determined by standard-of-care clinical intervals (e.g., routine restaging scans) **Remote Monitoring Frequency:** Continuous data extraction from selected healthcare databases **Checklist Review Interval:** Regular programmatic data refresh cycles

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## 15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- |
| **Principal Investigator** | [Name] | ____ **| [DD-MMM-YYYY] | |**

**Clinical Operations Lead | [Name] | ___ |** ***[DD-MMM-YYYY]*** **| |** ***Lead Statistician / Epidemiologist*** **|** ***[Name]*** **|** _____ | [DD-MMM-YYYY] |